

CALIMESA FIRE DEPARTMENT
POLICY MANUAL
VOLUME 6

Policy: 600.10 Structure Fires

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The purpose of the Structure Fires Policy is to provide a basic sequence of tasks required for the mitigation of structure firefighting.

ARRIVAL ON SCENE

1. The First Arriving Officer will provide a CURBSIDE scene size-up, assume command, establish the Alpha Side, and develop a fire suppression plan.
2. The Officer will complete or assign a 360-DEGREE viewing of the structure to determine the number of stories, type of structure, what is observed, location of problem, exposures, what is burning, where the fire appears to be going, any potential hazards, and the need for additional resources.
3. The Officer will ensure that a proper communications plan has been coordinated for fireground activities.
4. The first arriving officer shall declare the appropriate STRATEGY, MODE OF OPERATION and ACCOUNTABILITY.
5. The First Arriving Officer shall declare or request/assign a WATER SOURCE.

SCENE SAFETY

1. There shall be a minimum of 2 personnel on any interior attack hose line.
2. There shall be a minimum of 2 personnel on any interior search and rescue. If a RESCUE is declared the Officer declaring shall ensure the ECC and or responding Chief Officer or Next Arriving verbally acknowledges the radio declaration.
3. There should be a RIT/RIC or 2 - Out Standby in place on any interior operation or any situation where personnel are exposed or could be exposed to any IDLH situation.
4. Personnel must be authorized and properly trained before participating in any interior structural fire attack or search and rescue operations.

INCIDENT ACTIONS

1. Rescue
 - a. Human life is the most important consideration at a fire or other emergency.
 - b. Rescue of human life overrides all other strategic considerations at a fire.

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- c. The primary function of an adequately staffed truck company (if available) shall be rescue.
 - d. A primary and secondary search shall be conducted at all structure fires. During search each room or space should be marked by some means to indicate that each particular room has been satisfactorily searched.
- 2. Exposure Protection
 - a. Exposure protection is the strategy of preventing a fire from spreading to the uninvolved building(s) or in involved parts of the fire building.
 - b. The Incident Commander shall be responsible for ensuring the initial protection of exposures and assigning Groups or Divisions appropriately.
- 3. Confinement
 - a. The strategy of confinement means preventing the fire from extending to uninvolved sections of the building.
 - b. Whenever possible, the most effective method of confining fire spread is a direct attack on the fire.
 - c. The Incident Commander shall decide whether to make an offensive approach, aggressive interior attack or a defensive approach attacking the fire from the outside. There may be situations when both approaches could be used, but a defensive attack should not be used when crews are operating on the interior.
 - d. All avenues of fire spread must be considered examples: shafts, openings, utility raceways, ducts etc.
 - e. Where fires involve concealed spaces (attic, ceilings, construction voids, etc.) it becomes very important that the vent crews open, and fire attack operates fire streams into such areas.
- 4. Extinguishment
 - a. In most fire situations a quick and aggressive attack on the seat of the fire will take care of rescue, exposures, and confinement at the same time.
 - b. The size-up will provide information as to techniques, equipment and manpower needs to overcome the fire.
- 5. Overhaul
 - a. The purpose of overhaul is to make sure the fire is completely out.
 - b. Overhaul operations must be properly coordinated with fire investigation efforts.
 - c. Unsafe conditions should be identified early in the overhaul process and definite efforts made to avoid the possible problems associated with the same.
 - d. During overhaul most fire fighters are more relaxed, tired, perhaps less alert, and thus more apt to get injured.

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- e. Personnel should not remove their breathing apparatus until the area is completely cleared of toxic gases.
 - f. When available, a fresh crew should perform overhaul.
 - g. Particular attention should be given to hidden areas during overhaul.
 - h. During overhaul care should be given to protect personnel from exposure to carbon monoxide and other by products of combustion.
6. Ventilation
- a. Based upon the situation, ventilation may need to occur anytime during the operation.
 - b. Ventilation shall be employed to:
 - i. Channel heat, smoke, and flames from potential victims.
 - ii. To prevent backdraft and flashover.
 - iii. To remove heat and smoke from the building so to reduce property damage.
 - iv. To allow the interior of the structure to be more tenable and safer for firefighting operations.
7. Salvage
- a. Salvage may need to begin at various points during a fire operation.
 - b. Salvage is those operations required to safeguard personal property, furnishings, and the unaffected portions of a structure from the effects of heat, smoke, fire and the weather.
 - c. Salvage should include:
 - i. The use of salvage covers.
 - ii. Removing water from the structure.
 - iii. Removing furniture and personal belongings to a safe location.
 - iv. Debris removal.
 - v. Removal of valuables from debris.
 - vi. Covering openings to keep weather out and to secure the building.
 - d. All members are expected to perform in a manner that continually reduces loss during fire operations.
8. Utility Control
- a. Utilities should be shut down and brought under control to ensure that they will not contribute to the fires spread, overall damage or create any type of safety hazard.
 - b. At structure fires where electrical involvement or damage has occurred, request via radio the response of the proper electric company.

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- c. If the electric company is not available in time, fire personnel may shut down the power via circuit breakers.
 - d. If necessary, shut down gas lines at the meter and have the Gas Department notified.
 - e. Meters that have been shut off by fire department personnel should be properly locked.
 - f. If necessary, shut down water supplies to the structure at the valve closest to the point of usage.
9. Mid -Rise, High Rise, - Large “Big-Box” Commercial Fires shall require the required resources, accountability, ICS Positions, and adherence to the appropriate tactics

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This Policy is intended to serve as an operational template when operating in wildland firefighting operations.

OVERVIEW

The Wildland Urban Interface/Intermix (WUI) within the City of Calimesa and the surrounding communities occasionally experience vegetation fires with varied complexity and resource needs. Oftentimes these fires burn across, or threaten jurisdictional boundaries with Local (LRA), State (SRA), and Federal (FRA), responsibility Areas. These fires are typically responded to by agencies from within the Operational Area, with pre-determined Mutual Threat Zones (MTZ). As a standard of practice Local Government Agencies develop Automatic Aid Agreements and Participate in the California Master Mutual Aid System in cooperation with California Office of Emergency Services (CAL-OES). The Calimesa Fire Department shall utilize the concepts, tenants, and best practices as listed in documents and publications from California Fire Scope, National Wildfire Coordinating Group (NWCG), and CAL FIRE Wildland Urban Interface

DEFINITIONS

Air Attack - Aerial reconnaissance aircraft that can provide information from above the fire by an observer.

Air Tanker - Fixed wing aircraft certified by the FAA as being capable of transport and delivery of fire-retardant solutions.

Division - A geographic work assignment (example East Division, or Division A)

Group - A functional work assignment (example Water Supply Group)

LCES (Lookouts, Communications, Escape Routes, Safety Zones) - A basic safety measure that must be in place at all times for fire fighter safety.

Red Flag Warning - A weather announcement made by the National Weather Service when fire danger is Very High or Extreme. Sustained winds ≥ 20 mph and relative humidity $\leq 20\%$, scattered thunderstorms, increased thunderstorm activity after a prolonged dry period, abrupt change in wind speed and direction that may affect the area.

SEAT (Single Engine Air Tanker - similar to a crop duster) - A fixed wing aircraft capable of

transport and delivery of fire-retardant solutions of about 800 gallons.

Type 1 Engine - A typical structure engine. Crew of 4 personnel, minimum 400-gallon tank and 1000 gpm pump.

Type 3 Engine - Typically a short wheelbase engine capable of both structure and wildland fire operations. Commonly 4-wheel drive. Minimum 3 personnel, 500-gallon tank, 250-gpm pump.

Type 6 Engine - Typical “brush truck”. Minimum crew of two, approx. 200-gallon tank and 50 gpm pump with higher pump pressure capacity than an engine and pump and-roll capability. Typically, 4-wheel drive for off road fire suppression.

Water Tender (“Tender”) - A water transport and delivery fire apparatus. Capable of carrying 1000 to 5000 gallons of water. Pumps may range from 50 to 1500 gpm.

Wildland Fire - Any non-structure fire that occurs in the wildland.

GUIDELINE

Resources responding to a brush/wildland fire will be familiar with National Incident Management System (NIMS) terminology and practices. Resources should be prepared to work in Sector/Division/Group assignments or other positions with resources from outside the regional dispatch system.

Wildland fires typically demand a significant command and support staff to manage and will often require resources from other Fire Departments and Government agencies. Command will be responsible for requesting all needed resources.

FIRE FIGHTER SAFETY

PPE

In the brush/wildland fire setting proper personal protective equipment (PPE) is essential to fire fighter safety. Structural turnout coats, pants and boots are not designed for brush/wildland fire suppression. Wildland PPE is designed to be more comfortable and functional while reducing fire fighter fatigue and heat-related injuries.

A proper brush/wildland fire protective ensemble will consist of a helmet, fire resistive brush shirt/jacket, fire resistive brush pants, eye and hearing protection, work gloves and leather ankle high boots. Nylon hiking boots are inappropriate because of the melting and sticking potential of nylon.

HYDRATION

Remember that heat is a major safety problem, and all personnel should be kept well hydrated. Personnel should have access to drinking water and carry canteens or similar water containers. Sterile water bottles can be cleaned, filled with drinking water, and carried in the brush jacket

pockets.

Wildland fire fighting is a physically demanding operation and members should be fit and prepared mentally for a very hot, fast moving, and dangerous environment.

CREW SAFETY

Wildland fires demand that Captains maintain a high level of awareness regarding crew accountability. Crewmembers can easily become spread out and not visible in rugged and rocky terrain. Captains must maintain LCES (Lookouts, Communication, Escape Routes, and Safety Zones) and control over crewmembers to ensure a safe operation. Wildland fire fighting will still employ the buddy system. Watch out for each other.

TEN STANDARD FIRE ORDERS

Wildland fires are fast moving and extremely dangerous. These scenes require that all personnel understand these basic wildland fire-fighting orders:

1. Keep informed on fire weather conditions and forecasts.
2. Know what the fire is doing at all times.
3. Base all actions on current and expected behavior of the fire.
4. Identify escape routes and safety zones and make them known.
5. Post lookouts when there is possible danger.
6. Be alert. Keep calm. Act decisively.
7. Maintain prompt communications with your forces, supervisor, and adjoining forces.
8. Give clear instructions and insure they are understood.
9. Maintain control of your forces at all times.
10. Fight fire aggressively, having provided for safety first.

EIGHTEEN WATCH OUT SITUATIONS

Several situations have been found to increase the chance of injuries or fatalities on brush/wildland fires. These include:

1. Fire not scouted or sized-up.
2. In country not seen in daylight.
3. Safety zones and escape routes not identified.
4. Unfamiliar with weather and local factors influencing fire behavior.
5. Uninformed on strategy, tactics and hazards.
6. Instructions and assignments not clear.
7. No communications link with crewmembers/supervisor.
8. Constructing line without a safe anchor point.
9. Building fire line downhill with fire below.
10. Attempting a frontal assault on the fire.
11. Unburned fuel between you and the fire.
12. Cannot see the main fire or not in contact with anyone who can.

13. You are on a hillside where rolling material can ignite fuel below.
14. Weather is getting hotter and drier.
15. Wind increases and/or changes direction.
16. Getting frequent spot fires across the line.
17. Terrain and fuels make escape to safety zones difficult.
18. Taking a nap near the fire line.

FIRE BEHAVIOR

The following factors have a critical effect on the fire behavior of a wildland fire. Command must maintain an awareness of these conditions and be prepared to react quickly, pessimistically, and well ahead of the fire. The factors are weather, fuel, and topography.

WEATHER

Command must be aware of constantly changing weather conditions. During a normal day local wind will change 180 degrees near midday and usually become gusty during the afternoon. Morning winds are normally East to West and afternoon winds are usually West to East. Fire spread will usually slowdown in the evening as humidity increases (25%) and increase during the midmorning hours as the humidity decreases (15%).

A Red Flag Warning issued by the National Weather Service indicates when conditions are present that may have substantial effects on any brush/wildland fire incidents during that period. Crews should give special consideration to any fire incident under these conditions.

Command should always be aware of the fire conditions, weather conditions and time of day. Remember that a large WILDLAND fire can create dangerous convection currents that cause erratic fire behavior and spot fires far in advance of the fire head. Heavy winds also produce similar results.

Hot and dry conditions produce extremely rapid fire spread. A slight decrease in relative humidity will cause a significant increase in fire intensity. During extreme days' surface-wetted fuel will dry in a few minutes.

FUEL

Most of the fuel in the Calimesa area is relatively light and burns very rapidly. It is not expected to burn in any area for more than a few minutes and may not require extensive overhaul.

TOPOGRAPHY

Fire burns uphill much more rapidly than downhill. On an uphill slope the fire will tend to crown over the top and start spot fires a considerable distance down the receding slope. A large free-

burning fire will tend to create its own convection currents and spot fires may be started. Access is often the most serious problem with topography.

Companies with considerable brush fire potential should size-up areas with regard to fuel, topography and extent of exposure to structures. Attention should be paid to access roads and accessible areas where apparatus may travel. Natural firebreaks and potential exposure problems should be noted on the area maps provided for this purpose.

COMMAND

The first arriving company officer that assumes Command must address the life safety, fire control, and property conservation benchmarks. In a wildland fire setting the life safety benchmarks must include fire fighters as well as civilians. Command must have a plan that includes LCES (Lookouts, Communications, Escape Routes and Safety Zones) for fire fighters and equipment. LCES should be in place prior to any fire suppression operations. Individual sectors can establish escape routes and safety zones depending on need or location. Escape routes and safety zones should be easily accessible and large enough to prevent radiant heat injuries or direct flame impingement. Aerial operations should not be located near safety zones.

- Command should concern itself with strategy and allow sectors to dictate tactics.
- Make early offensive/defensive (direct attack or indirect attack) decisions.
- Gather adequate resources
- Think ahead - way ahead.
- Support sectors
- Protect and rehab fire fighters.
- Consider filling the operations chief and/or logistics chief position early into the incident!

On major incidents establish a Command Post and Divisions/Groups as soon as possible. The Command Post should be in a location that will be safe and not have to move if the fire changes direction. Supporting elements should be able to set up at the Command Post. Select a site where a helicopter may land in close proximity to the Command Post.

The Command Post will need to plot progress, exposures and access. The units in the field, particularly geographic sectors/divisions/groups, will have to report this information back to Command via radio or personal contact.

As fire spread becomes critical Command must be prepared to special call additional attack units by specific companies or to request assistance by standardized alarm responses. This determination must be made early.

When brush/wildland fires begin to threaten homes Command must be prepared to readjust and develop a defensive strategy to protect exposures while allowing the fire to burn to a location better suited for control.

TACTICS AND STRATEGY

Brush fires often present a large area of rapidly spreading fire. The critical decision is often where to attack the fire to the best advantage. Protection of exposure is the primary goal when immediate control is not possible.

SIZE UP

A deliberate and thorough initial size up is extremely important on significant or potentially significant fire incidents in order to develop a firefighting plan. The initial incident commander should take the necessary time to gather pertinent information on the fire's location, spread rates and direction, identified hazards, any threats to improvements (structures, power lines, etc.) and anticipated needed resources.

The following is a list of size-up considerations that greatly affect tactics and strategy.

- Location of fire head or heads.
- Pertinent burning conditions: weather, time of day, etc.
- Type of fuel: light, heavy fuel.
- Exposures: improvements, buildings, crops, etc.
- Size of fire and rate of speed.
- Special hazards: spot fires, hazardous materials
- Manpower needs.
- Fuel continuity.
- Accessibility into fire area.
- Water resources - tenders, hydrants, etc.
- Line of retreat. How can I escape?

Command must then quickly develop an incident action plan based on this size up.

DIRECT ATTACK (OFFENSIVE)

Direct attack should be used whenever fire conditions allow fire personnel to work directly and safely on the fire's edge. Personnel should "anchor and flank" a fire by first establishing a safe location, or anchor point, to start the attack without being outflanked by fire. An example of an anchor point would be a road or green farm field.

Fire suppression personnel on fires in light fuels should use the "one foot in the black" method in an inside out attack where the safety zone is the previously burned area adjacent to the burning fire front. Unlike a structural fire attack, a brush/wildland fire attack should be from the burned side where possible.

Structural fire fighters can find themselves entrapped by fire when attacking a fire head-on from the unburned side where fire can quickly overrun them.

A direct water attack is the fastest control evolution available to counteract wildfire spread. Brush

trucks can accomplish this through pump-and-roll tactics. Apparatus and personnel should be in the burned area as opposed to the unburned fuels.

INDIRECT ATTACK (DEFENSIVE)

Indirect attack methods are used when fire personnel are prohibited from direct attack due to fire conditions or access to the fire. For indirect attack strategy fire personnel work some distance away from the fire's edge. This may be in support of wildland fire crew burnout operations, structure protection or another tactic.

On large open grass fires Command must take advantage of natural fire barriers that will assist in control measures such as: dry sandy washes, roads, trails, rock outcroppings, patch fuels, etc.

STRUCTURE DEFENCE INTERFACE OPERATIONS

During interface fires involving brush and structures it may be necessary to retain a 3-1 structural assignment in staging and be prepared to assign those units to conduct structural firefighting if needed. These staged resources will be turned out and equipped to carry out interior structural firefighting.

Units assigned to protect a structure or improvement should first thoroughly size up the site to ensure fire fighter safety can be maintained. The company officer should walk the driveway or access road to ensure apparatus have good access and clearance and that the site is deemed safe to protect. Apparatus should back in to allow for quick egress if necessary. During structure protection crews should plan to remain as mobile as possible in case escape is necessary. Hose lays should be as short as possible and limited to one or two at most. Some regional departments that have more frequent or substantial interface areas may use "Bump bags" or other hose packs. These hose packs consist of 1 ½" "trunk" line with a gated "y" and one or two 1" forestry hose lines with nozzle.

When water is in short supply it is usually most effective when applied to burning material instead of wetting fuel in advance. Seriously exposed structures should be kept wet.

Tactical challenges and hazards for structure protection:

(Firefighters with a safety zone can safely defend structures with some challenges)

- Narrow roads, unknown bridge limits, and septic tank locations.
- Ornamental plants and combustible debris next to structure.
- Wooden siding and/or wooden roof materials.
- Open roof vents, eaves, decks, and other ember traps.

- Fuel tanks and hazardous materials.
- Power lines • Limited water sources.
- Property owners remaining onsite.

MOP-UP

After direct or indirect line work is completed and a fire is called "Contained" many things remain to be done to make the fire line safe and put the fire out. This work is called mop up. The objective of mop up is to put out all fire embers or sparks to prevent them from crossing the fire line.

A certain amount of mop up work is done along with line building. Mop up becomes an independent part of firefighting as soon as the spread of the fire is stopped and all line has been completed. Ordinarily mop up is composed of two actions; putting the fire out and disposing of fuel either by burning to eliminate it, or removing the fuel so it cannot burn.

The principles of mop up follow:

- Start work on each position of line just as soon as possible after line construction and burning out are completed. Treat most threatening situations first.
- Allow fuel to burn up if it will do so promptly and safely.
- On small fires, all fire should be extinguished in the mop up, where quantities of burning material are not so large as to make this impractical.
- On large fires, completely mop up enough of the area adjacent to the line to be certain no fire can blow, spot, or roll over the fire line under the worst possible conditions.
- Search for smoldering spot fires.
- All smoldering material that is not put out with water or dirt should be spread well inside of lines.
- Eliminate or put into a safe area all less flammable fuels, such as rotten logs and snags, which are outside, but near the control line.
- Eliminate all burned trees inside of line that could throw sparks over line or fall over the line.
- Put all rolling material in a position that it cannot possibly roll across the line.
- Look for indications of hot spots. Some are gnats swarming, white ash, ground which shows pinholes, and wood boring insects.
- Use water wherever possible and practical in mop up.
- Use water sparingly, but use enough to do the job. Match the amount of water to the job.
- Adding Class A foam to water will greatly increase effectiveness in mop up of deep-burning fuels.

When addressing mop up operations Command should:

- Determine the distance inside the control line to be overhauled (for small fires, this may be the entire burn area).
- During rehab of mop up crews ensure at least two fire fighters remain in the area to monitor for re-ignition or spread of fire.

- Schedule for follow-up checks by crews to ensure the fire is out in mopped up perimeter